CMAN :=

Given data, $N=32$, FP32 = 4 bytes, DRAM bandwidth = 320GB/s
compute = 10TFLOPS.

Task 1 :=

N=32 times, so each B[K][J] accessed 32 times

A access $N^3 = 32^3 = 32{,}768$

B access $N^3 = 32^3 = 32{,}768$

Total elements access $= 2N^3 = 2\times(32)^3 = 65{,}536$

Total DRAM units $= 2N^3 \times 4 = 65{,}536 \times 4$

$= 262{,}144$ bytes $\simeq 256$KB

$\boxed{\text{Total DRAM traffic} = 256\text{KB}}$

Task 2 :=

Tile size (T=8)

Number of tiles per dimension $= N/T = \frac{32}{8} = 4$

A Tile loads the each A&B tiles used N/T times
(once per tile column of B)

A loads $= \left(\frac{N}{T}\right)^3 \times T^2 = (4)^3 \times 8^2 = 64 \times 64 = 4096$ elements

B tile also same = 4096 elements

Total DRAM traffic $= 2 \times 4096 \times 4 = 32{,}768$ bytes

$= 32$KB

$\boxed{\text{Total DRAM traffic} = 32\text{KB}}$

Task 3 :=

$$\text{Traffic ratio} = \frac{\text{Naive}}{\text{Tiled}} = \frac{262,144}{32,768} = 8\ (T)$$

each TxT tile of B is loaded from DRAM and then reused accross T output tile rows before eviction so tiling reduces total DRAM traffic by exactly a factor of T and since the same reuses applies symmetrically to A. the overall naive to tiled traffic ratio equals the title size T=8. is not equal to N. The ratio only equals N when T=N (full caching).

for T=8, N=32 → ratio = T=8.

Task 4 :=

BW = 320 GB/s

Compute = 10 TFLOPS

Total FLOPS $= 2 \times N^3 = 2 \times 32^3 = 65,536$ FLOPS

Naive case :=

$$t_memory = \frac{\text{Traffic}}{BW}$$

$$= \frac{262,144}{320\,GB/s}$$

$$= 8.192 \times 10^{-7} s = 0.819\ \mu s$$

$$\boxed{t_memory = 0.819\ \mu s}$$

$t_compute = FLOPs / P$

$= 65{,}536 / (1 \times 10^{13})$

$= 6.554 \times 10^{-9} s = 0.00655 \mu s$

$t_memory\ (0.819 \mu s) >> t_compute\ (0.00655 \mu s)$

$\boxed{t_exec\ (naive) = 0.819 \mu s}$

Bottleneck: Memory-bound.

Tiled case (T=8):=

$t_Memory = Traffic / BW$

$= 32{,}768 / (3.2 \times 10^{11})$

$= 1024 \times 10^{-7} s = 0.1024 \mu s$

$t_compute = FLOPs / P$

$= 65{,}536 / (1.0 \times 10^{13})$

$= 6.554 \times 10^{-9} s = 0.00655 \mu s$

$t_memory\ (0.1024 \mu s) >> t_compute\ (0.00655 \mu s)$

$\boxed{t_exec\ (tiled) = 0.1024 \mu s}$

Bottleneck: Memory-bound